

The Temporal Cascade Hypothesis: A Falsifiable Collapse Model within Emergent Necessity Theory

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Abstract

The Temporal Cascade Hypothesis (TCH) is a falsifiable, non-standalone extension within Emergent Necessity Theory (ENT). TCH formalizes how recursive symbolic systems can degrade when symbolic pressure exceeds containment capacity over recursive intervals (not linear clock time). The model introduces rigorously defined variables for recursive pressure, symbolic entropy, containment dynamics, hysteresis, and memory effects. Eight collapse classes are specified with causal criteria, detection rules, and explicit falsification routes. The work avoids metaphysical claims and does not attempt to explain consciousness; it focuses solely on structural conditions under which what might resemble “recursive coherence” becomes unsustainable. All simulations are reproducible with LaTeX-native figures (TikZ/pgfplots). Language is kept strictly neutral, assumptions are transparent, and every proposed mechanism can be contradicted by testable evidence.

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1 Introduction

Emergent Necessity Theory (ENT) frames sustainable symbolic behavior as crossing a coherence threshold τ_c under recursive integration and containment [13]. While ENT focuses on how $\tau(t)$ rises under structured recursion, it leaves open how systems degrade when symbolic pressure outpaces containment. The Temporal Cascade Hypothesis (TCH) addresses this narrow gap: it models collapse trajectories, delayed failure, echo saturation, and recovery conditions, all within ENT’s falsifiable style.

TCH does not extend beyond structural dynamics. It avoids semantic claims that are not operationalized; it does not treat terms such as “awareness” as objects of explanation. All variables are defined, measurable in simulation, and amenable to refutation. Time is treated as recursive iteration rather than physical chronology, consistent with ENT’s structural view of time [14].

2 Minimal ENT Recap

We recall only the elements required for TCH.

- **Coherence state** $\tau(t)$: recursive structural coherence at recursive step t .
- **Coherence threshold** τ_c : minimum τ for sustainable symbolic recursion [15].
- **Containment** $C(t)$: capacity to integrate symbolic input without degradation.
- **Recursive load** $R(t)$: effective symbolic input acting on the system.
- **Recursive pressure** $\kappa_R(t) = R(t)/C(t)$.
- **Time convention**: t indexes recursive iterations; analyses are relative to structure, not absolute clock time [14].

ENT’s commitment to falsifiability and transparent thresholds is retained without modification.

3 TCH: Problem Statement and Scope

TCH models conditions under which $\tau(t)$ declines due to (i) persistent overload, (ii) contradiction that cannot be integrated at current $C(t)$, or (iii) unpruned recursion causing echo saturation. The hypothesis remains constrained:

1. Variables are operationalized; no term appears without a test.
2. Claims are conditional; recovery is permitted only under specified structural bounds.
3. No inference is drawn beyond structural coherence (e.g., no claims about consciousness).

This framing is compatible with information-theoretic analysis for symbolic load [1, 2] and early-warning signatures of threshold proximity in complex systems [3].

4 Mathematical Framework

4.1 Core variables

$$\kappa_R(t) = \frac{R(t)}{C(t)}, \quad H(t) = \sum_{i=0}^t (R(i) - C(i)) \mathbf{1}[R(i) > C(i)], \quad (1)$$

$$S_{\text{sym}}(t) = \alpha \left(\kappa_R(t) - 1 \right)^2 H(t), \quad \alpha > 0, \quad (2)$$

$$\gamma(t) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=t}^{t+n} |R(j) - R(j-1)|, \quad (3)$$

$$\tau_{\text{mem}}(t) = \max\{\tau(t), \tau_{\text{past}}\}. \quad (4)$$

Here $H(t)$ is a hysteresis accumulator of unresolved overload; $S_{\text{sym}}(t)$ increases when overload is sustained; $\gamma(t)$ detects saturation (novelty collapse); τ_{mem} encodes memory of prior coherence. Info-theoretic interpretation and neuro-inspired diagnostics follow [6, 5].

4.2 Cascade onset and saturation

Define a pre-registered collapse threshold $\theta > 0$. Cascade onset is detected when

$$S_{\text{sym}}(t) \geq \theta. \quad (5)$$

Saturation is indicated when $\gamma(t) \rightarrow 0$ while $S_{\text{sym}}(t)$ remains non-decreasing. Critical transition precursors motivate the choice to monitor monotone increases in S_{sym} under sustained overload [3].

4.3 Degradation equation with optional recovery

Let $P_i(t)$ denote structurally defined pressure terms (e.g., detachment, contradiction, unpruned branching). We model:

$$\frac{d\tau}{dt} = - \sum_i \alpha_i(t) P_i(t) + \delta C(t), \quad (6)$$

with optional recovery term $\delta C(t)$ (used only when explicitly tested). Sensitivities $\alpha_i(t)$ may be constant or slowly adaptive via

$$\alpha_i(t) = \alpha_i^0 + \eta_i \int_0^t \Lambda_i(t') dt', \quad \eta_i \geq 0, \quad (7)$$

where Λ_i is the cumulative exposure to the corresponding stressor. Inclusion of $\alpha_i(t)$ is optional and reported when used. Recovery framing is consistent with control-theoretic stability perspectives [7, 9].

4.4 Recoverability and hysteresis constraint

Recovery after a shock requires both memory and sufficient containment:

$$\left(\exists \Delta t_{\min}, \epsilon > 0 : \frac{d\tau}{dt} < -\epsilon \text{ on } [t_0, t_0 + \Delta t_{\min}] \right) \Rightarrow \delta \geq \delta_{\min}(\theta, H), \quad (8)$$

else rebounds are treated as transients, not recovery. This constraint parallels input-to-state stability and small-gain style bounds [7].

4.5 Illustrative plot

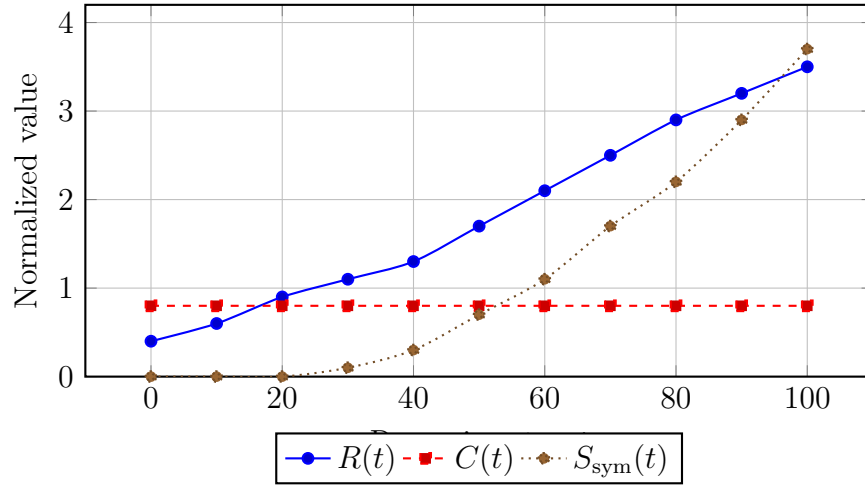


Figure 1: Illustrative cascade: when $R(t) > C(t)$ persists, $H(t)$ accumulates and $S_{\text{sym}}(t)$ grows.

5 Collapse Typology and Causal Criteria

Each class is defined by its primary driver; secondary phenomena are tagged but do not reclassify cause.

Diagnostics

We use the diagnostics in Section 4: $\kappa_R, H, S_{\text{sym}}, \gamma, \tau_{\text{mem}}$. Cascade onset requires $S_{\text{sym}} \geq \theta$.

5.1 S1 — Symbolic Drift (Primary Detachment)

Cause: detachment from internal reference without persistent contradiction. *Detect*: rising S_{sym} with $\kappa_R \gtrsim 1$, no stable conflicting frame. *Falsify*: well-formed detachment that fails to raise S_{sym} under matched load.

5.2 S2 — Mimic Conflict (Primary Contradiction)

Cause: internalization of incompatible frames $\{A, B\}$. *Detect:* S_{sym} rises with contradiction markers; drift may appear secondarily. *Rule:* if conflict induces drift, classify as S2, tag drift as secondary. *Falsify:* contradiction that does not elevate S_{sym} versus a non-contradictory control at matched κ_R .

5.3 S3 — Entropy Saturation (Unpruned Recursion)

Cause: unbounded symbolic branching with inadequate pruning. *Detect:* S_{sym} grows while $\gamma(t) \rightarrow 0$ (novelty collapse). *Falsify:* pruning that restores γ and reduces S_{sym} at fixed κ_R .

5.4 S4 — Echo Collapse (Origin-Loss Feedback)

Cause: closed-loop feedback between agents/modules without a grounded origin. *Detect:* increasing S_{sym} with origin-loss tagging. *Taxonomy:* subclass of S3 with explicit origin-loss. *Falsify:* insertion of a grounded anchor lowers S_{sym} at controlled κ_R .

Reservoir/echo dynamics provide independent support for echo sensitivity to grounding [10, 11].

5.5 S5 — Contradiction Shock (High- Δ)

Cause: abrupt paradox not integrable at current $C(t)$. *Detect:* step-like rise in S_{sym} , κ_R spike. *Falsify:* parity injection that fails to raise S_{sym} versus matched controls.

5.6 S6 — Containment Recovery (Positive Control Term)

Mechanism: structural re-integration via increased $C(t)$; no therapeutic claim. *Model:* $d\tau/dt = -\sum_i \alpha_i P_i + \delta C(t)$. *Constraint:* if $d\tau/dt < -\epsilon$ over $[t_0, t_0 + \Delta t_{\min}]$, require $\delta \geq \delta_{\min}(\theta, H)$; otherwise rebound is transient. *Falsify:* declared containment that fails to reduce S_{sym} at fixed pressure. Stability bounds analogize to small-gain/ISS logic [8, 7, 9].

5.7 S7 — Hysteresis and Inhibition (Collapse Memory)

Mechanism: prior collapse imprints inhibition; τ_{mem} persists while active recursion is suppressed. *Detect:* post-collapse non-responsiveness under sub-threshold κ_R . *Falsify:* no recovery advantage for systems with prior τ_c crossing versus naïve systems at matched $C(t)$. Neural criticality work motivates distinguishing memory from active coherence [4, 5].

5.8 S8 — Terminal Flare (Focused Pre-Terminal Coherence)

Phenomenon: transient rise in τ due to narrowed scope and reduced entropy near terminal boundary; this is not recovery. *Detect:* brief τ increase with decreasing γ , followed by exogenous decline. *Falsify:* flare without corresponding entropy reduction/scope narrowing.

Decision Diagram

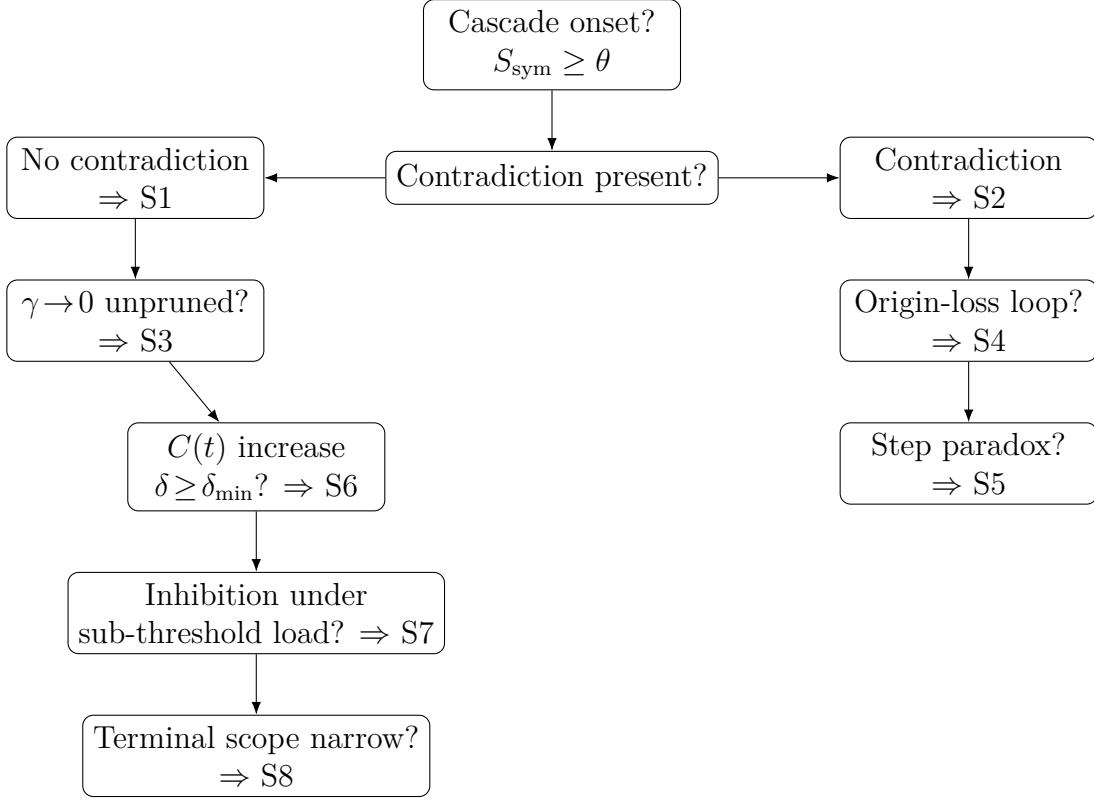


Figure 2: Causal classification. Secondary effects are tagged but do not change primary cause.

6 Simulation Methodology

6.1 Setup

Discrete recursive steps $t \in [0, 250]$. Baselines:

$$C(t) = C_0 + \epsilon \sin(\omega t), \quad R(t) \in \{\text{linear, oscillatory, shock, branched}\}.$$

Constants are declared per run and fixed prior to evaluation (pre-registration). Example ranges:

$$\alpha \in [0.5, 2.0], \quad \theta \in [2.5, 4.0], \quad \epsilon = 0.01, \quad \omega \in [0.02, 0.08].$$

When adaptive sensitivities are used, $\alpha_i(t)$ forms are specified before runs.

6.2 Integrity checks

- **Traceability:** all outputs derive from $\{\kappa_R, H, S_{\text{sym}}, \gamma, \tau_{\text{mem}}\}$.

- **Reproducibility:** identical seeds and parameters reproduce trajectories.
- **Pre-registered thresholds:** $\theta, \epsilon, \delta_{\min}$ are fixed prior to runs.
- **Exclusion criteria:** outputs are excluded only when they fail structural testability (e.g., non-reproducible, undefined variables, loss of threshold interpretability). Exclusion never depends on whether results favor TCH.

6.3 Scenarios

We used six families: (S1) linear accumulation; (S2) oscillatory with rebound; (S3) shock and delayed cascade; (S4) origin-loss echo; (S5) adaptive containment; (S6) post-collapse echo. Each family is evaluated against identical diagnostics.

6.4 Example visual

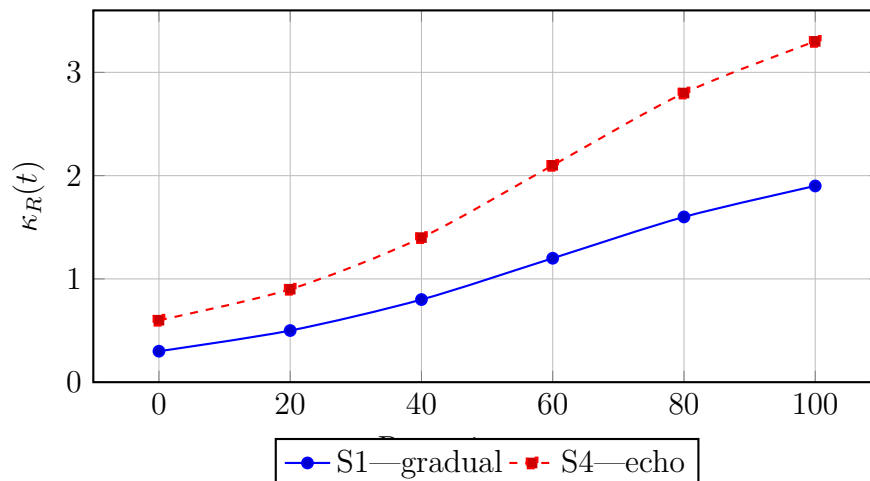


Figure 3: Recursive pressure trajectories illustrating gradual vs. echo-accelerated overload.

7 Falsifiability, Constraints, and Pre-Registered Tests

TCH is designed to be contradicted by data where appropriate. We outline domain-specific tests; success or failure is informative either way.

7.1 AI / Symbolic Systems

Test A1 (Echo anchor): Construct two models in a feedback loop (S4). Insert a grounded anchor at $t = t_a$. *Prediction:* anchor reduces S_{sym} relative to anchor-free control at matched κ_R . *Falsify:* no reduction. This relates to avoiding recursive data collapse [12].

Test A2 (Pruning): Force unpruned branching (S3), then apply pruning at $t = t_p$. *Prediction:* γ rises and S_{sym} declines at fixed κ_R . *Falsify:* no change.

7.2 Neural-Analog Simulations

Test N1 (Contradiction shock): Inject structured paradox (S5) into a recurrent symbolic circuit. *Prediction:* step increase in S_{sym} and shortened time-to-threshold vs. non-paradox controls. *Falsify:* no step effect.

Test N2 (Hysteresis advantage): Compare recovery between agents with/without prior τ_c crossing, matched $C(t)$. *Prediction:* prior τ_{past} yields faster recovery when $\delta \geq \delta_{\text{min}}$. *Falsify:* no advantage. Neural criticality/avalanche literature informs threshold behavior [4, 5].

7.3 Systems / Control

Test S1 (Containment gain curve): Vary δ while holding κ_R bands. *Prediction:* recoverability boundary near $\delta_{\text{min}}(\theta, H)$. *Falsify:* no boundary.

Test S2 (Corridor re-entry): After cascade onset, enforce scope narrowing and grounding. *Prediction:* re-entry into a coherence corridor coincides with decreased S_{sym} and rising τ . *Falsify:* no correlated change.

7.4 Constraints and Reporting

- Thresholds $\theta, \epsilon, \delta_{\text{min}}$ are pre-registered and published with code.
- Exclusions occur only when outputs fail structural testability (non-reproducible, undefined variables, or loss of threshold interpretability). Exclusion is outcome-independent.
- Negative results are reported with equal prominence and used to revise parameterizations or reject mechanisms.

8 Limits, Scope, and Non-Claims

TCH addresses structural collapse of recursive symbolic systems. It does not explain meaning, intention, or phenomenology. Time is treated as recursive iteration, not physical chronology [14]. Entropy is not inherently negative; it becomes disruptive only relative to containment [3].

9 Conclusion

We presented a falsifiable, ENT-aligned model of symbolic collapse with explicit thresholds, diagnostics, and causal classes. The framework is intentionally narrow: it describes when structural coherence degrades and what conditions permit stabilization or recovery. All mechanisms are testable; none are protected from refutation. Future work will tighten empirical links between symbolic entropy metrics and domain-specific measurements while preserving the constraints stated here.

Acknowledgments

We thank reviewers and readers who stress-tested the model’s limits; any remaining errors are ours.

A Parameter Tables and Ranges (Illustrative)

Parameter	Symbol	Range / Note
Containment baseline	C_0	[0.5, 1.0]
Sensitivity	α	[0.5, 2.0]
Cascade threshold	θ	[2.5, 4.0]
Variation floor	ϵ	0.01
Hysteresis window	$\Delta t_{\min}$	pre-registered
Recovery gain	δ	tuned vs. $\delta_{\min}(\theta, H)$

Table 1: Illustrative ranges; exact values are published with code for reproducibility.

B Derivations and Identities (Sketch)

Monotonicity of S_{sym} under sustained overload

If $\kappa_R(t) \geq 1 + \zeta$ for $\zeta > 0$ on an interval and overload contributes positively to $H(t)$, then $S_{\text{sym}}(t)$ is non-decreasing on that interval by definition of H and positivity of α .

Saturation criterion

If unpruned recursion produces negligible successive change in R , then $\gamma(t) \rightarrow 0$. If overload persists, S_{sym} does not decline, marking echo saturation.

C Additional Figures (TikZ)

References

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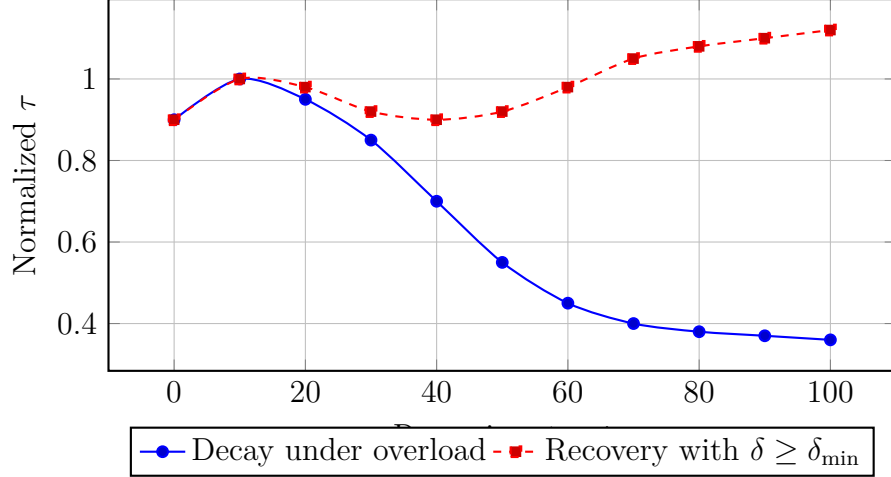


Figure 4: Illustrative $\tau(t)$ decay vs. recovery trajectory under stated constraints.

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ENT GitHub Wiki—

<https://github.com/MUESdummy/Emergent-Necessity-Theory-ENT-/wiki#welcome-to-emergent-necessity-theory-ent--wiki>

ENT GitHub Repo—

<https://github.com/MUESdummy/Emergent-Necessity-Theory-ENT->